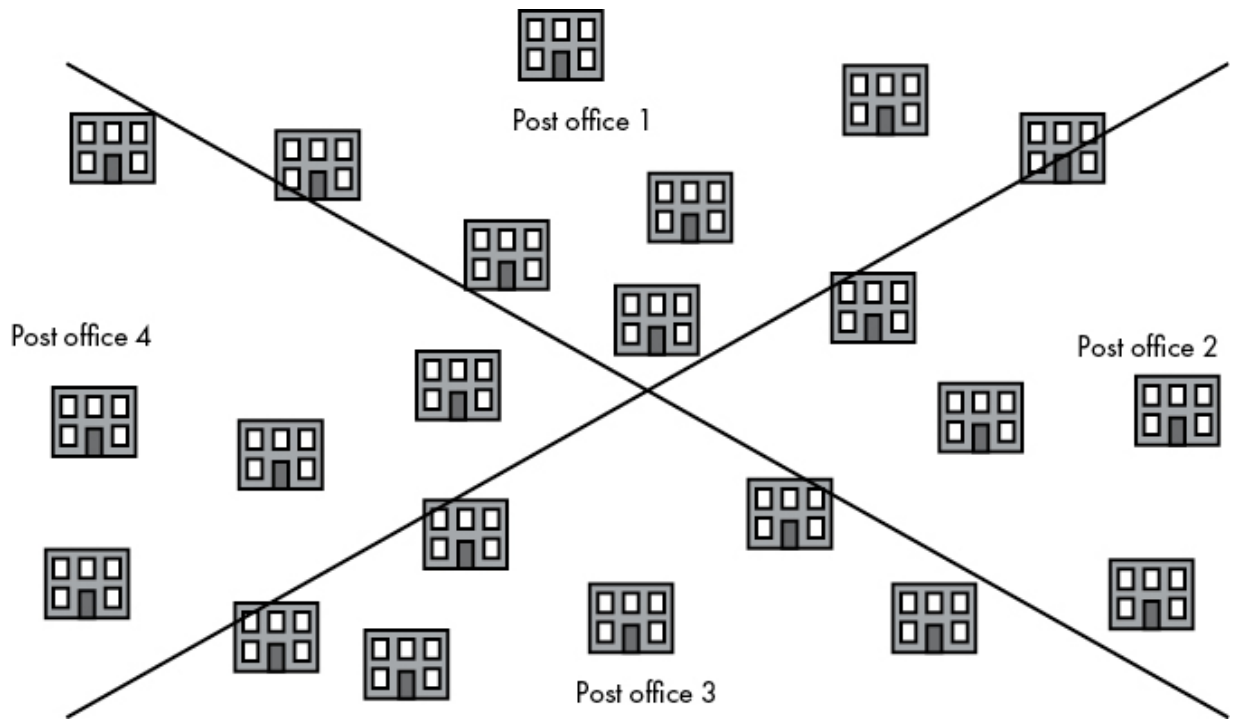




*Figure 7-3: Voronoi diagram separating our town into optimal postal delivery regions*

## Triangles 101

Let's back up and start with the simplest elements of the algorithms we'll explore. We're working in geometry, in which the simplest element of analysis is the point. We'll represent points as lists with two elements: an x-coordinate and a y-coordinate, like the following example:

---

```
point = [0.2,0.8]
```

---

At the next level of complexity, we combine points to form triangles. We'll represent a triangle as a list of three points:

---

```
triangle = [[0.2,0.8],[0.5,0.2],[0.8,0.7]]
```

---

Let's also define a helper function that can convert a set of three disparate points into a triangle. All this little function does is collect three points into a list and return the list:

---

```
def points_to_triangle(point1,point2,point3):  
    triangle = [list(point1),list(point2),list(point3)]
```

---